Exam 4. Practice 2.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

Notice: For all the maximization problems, it is required to do the three-step process to solve the problem.

1. Find $f'(x)$. Simplify 1c.
   a. $f(x) = 2x^7 - 8x + 9$

   $f'(x) = 14x^6 - 8$

   b. $f(x) = \frac{3}{x^3} + \frac{4}{x^3} + \frac{5}{x} + 3x + 1$

   $f(x) = 3x^{-3} + 4x^{-3} + 5x^{-1} + 3x + 1$

   $f'(x) = -9x^{-4} - 12x^{-4} - 5x^{-2} + 3$

   c. $f(x) = \frac{3x \quad +1}{4x \quad +1}$

   $f'(x) = \frac{(3x+1)' \cdot (4x+1) - (3x+1) \cdot (4x+1)'}{(4x+1)^2}$

   $f'(x) = \frac{3(4x+1) - (3x+1)4}{(4x+1)^2}$

   $f'(x) = \frac{-9}{(4x+1)^2}$

   d. $f(x) = (x^2 + 1)(x^{13} + x)$

   $f'(x) = (2x) \cdot (x^{13} + x) + (x^2 + 1)(13x^{12} + 1)$

2.

The per-day cost function (in dollars) for manufacturing a specific high-end electronic component is given by:

$$C(x) = 108,000 + 40x + 2x^3$$

Where x is the number of units manufactured per day ($0 < x \leq 100$).

a. Find the Marginal Cost function.
b. Calculate and interpret the meaning of $C'(20)$.
c. Write the Average Cost function, $AC(x)$.
d. How many items should be manufactured per day in order to minimize the average cost?

a. Find the Marginal Cost function.

The Marginal Cost (MC) is the derivative of the total cost function, denoted as $C'(x)$. Using the power rule:

$$C'(x) = (108,000)' + (40x)' + (2x^3)'$$

$$C'(x) = 0 + 40 + (3 \cdot 2)x^{3-1}$$

Answer: $C'(x) = 40 + 6x^2$

b. Calculate and interpret the meaning of $C'(20)$.

Substitute $x = 20$ into the marginal cost function:

$$C'(20) = 40 + 6(20)^2$$

$$C'(20) = 40 + 6(400)$$

$$C'(20) = 40 + 2,400$$

Result: $C'(20) = 2,440$

Interpretation: When 20 units are being manufactured per day, the cost is increasing at a rate of **$2,440 per unit**. This approximately represents the cost of producing the next (21st) unit.

c. Write the Average Cost function, $AC(x)$.

To find the Average Cost, divide the total cost by the number of units x:

$$AC(x) = \frac{C(x)}{x} = \frac{108,000 + 40x + 2x^3}{x}$$

Answer: $AC(x) = \frac{108,000}{x} + 40 + 2x^2$

d. How many items should be manufactured per day in order to minimize the average cost?

Step 1: Find $AC'(x)$

Start with the average cost function from part **c**:

$$AC(x) = 108,000x^{-1} + 40 + 2x^2$$

Using the power rule:

$$AC'(x) = -108,000x^{-2} + 0 + 4x$$

$$AC'(x) = -\frac{108,000}{x^2} + 4x$$

Step 2: Solve $AC'(x) = 0$

Set the first derivative to zero to find the value of x:

$$-\frac{108,000}{x^2} + 4x = 0$$

$$4x = \frac{108,000}{x^2}$$

$$4x^3 = 108,000$$

$$x^3 = 27,000$$

$$x = \sqrt[3]{27,000}$$

$$x = 30$$

Step 3: Verify that $AC''(x) > 0$

Find the second derivative to confirm that $x = 30$ results in a minimum:

$$AC''(x) = (-2)(-108,000)x^{-3} + 4$$

$$AC''(x) = \frac{216,000}{x^3} + 4$$

Now, substitute $x = 30$ into the second derivative:

$$AC''(30) = \frac{216,000}{(30)^3} + 4$$

$$AC''(30) = \frac{216,000}{27,000} + 4$$

$$AC''(30) = 8 + 4 = 12$$

Since $12 > 0$, the function is concave up at $x = 30$, confirming that the average cost is at its **minimum** when **30 items** are manufactured per day.

3.

A professional pickleball paddle manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity x of paddles they produce by the following quadratic demand function:

$$p(x) = -x^2 + 90x + 4,800$$

a. Find the revenue function $R(x)$ and the marginal revenue $MR(x)$ function.

b. Calculate and interpret the meaning of $MR(20)$.

c. Find the quantity x that should be produced in order to maximize the revenue function.

a. Find the revenue function $R(x)$ and the marginal revenue $MR(x)$ function.

The **Revenue function** is found by multiplying the price $p(x)$ by the quantity x:

$$R(x) = x \cdot p(x) = x(-x^2 + 90x + 4,800)$$

Answer: $R(x) = -x^3 + 90x^2 + 4,800x$

The **Marginal Revenue function** is the derivative of the revenue function, $R'(x)$:

$$MR(x) = R'(x) = (-x^3)' + (90x^2)' + (4,800x)'$$

Answer: $MR(x) = -3x^2 + 180x + 4,800$

b. Calculate and interpret the meaning of $MR(20)$.

Substitute $x = 20$ into the marginal revenue function:

$$MR(20) = -3(20)^2 + 180(20) + 4,800$$

$$MR(20) = -3(400) + 3,600 + 4,800$$

$$MR(20) = -1,200 + 3,600 + 4,800 = 7,200$$

Interpretation: When 20 units are produced and sold per day, the revenue is increasing at a rate of **$7,200 per unit**. This represents the approximate additional revenue generated by selling the **21st unit**.

c. Find the quantity x that should be produced in order to maximize the revenue function.

To maximize revenue, find where the first derivative is zero and verify with the second derivative.

Step 1: Find $R'(x)$ (Already found in part a):

$$R'(x) = -3x^2 + 180x + 4,800$$

Step 2: Solve $R'(x) = 0$
Set the marginal revenue to zero:

$$-3x^2 + 180x + 4,800 = 0$$

Divide the entire equation by -3 to simplify:

$$x^2 - 60x - 1,600 = 0$$

Factor the quadratic:

$$(x - 80)(x + 20) = 0$$

Since production quantity must be positive, we take $x = 80$.

Step 3: Verify that $R''(x) < 0$
Find the second derivative to confirm that $x = 80$ results in a maximum:

$$R''(x) = -6x + 180$$

Substitute $x = 80$:

$$R''(80) = -6(80) + 180 = -300$$

Since $-300 < 0$, it confirms that the revenue is at its **maximum** when **80 paddles** are produced per day.

4.

A company determines that when x units of a product are produced each month, they will be sold at a price of:

$$p = 60 - 3x$$

dollars per unit. The total cost of producing the x units is:

$$C(x) = x^2 + 12x + 40$$

a. Find the profit function $P(x)$ and the marginal profit $MP(x)$ function.
b. Calculate and interpret the meaning of $MP(5)$.
c. How many units should the company produce to maximize the profit?

a. Find the profit function $P(x)$ and the marginal profit $MP(x)$ function.

First, find the **Revenue function**, $R(x)$:

$$R(x) = x \cdot p(x) = x(60 - 3x) = 60x - 3x^2$$

The **Profit function**, $P(x)$, is Revenue minus Cost:

$$P(x) = R(x) - C(x)$$

$$P(x) = (60x - 3x^2) - (x^2 + 12x + 40)$$

$$P(x) = 60x - 3x^2 - x^2 - 12x - 40$$

Answer: $P(x) = -4x^2 + 48x - 40$

The **Marginal Profit function** is the derivative of the profit function, $P'(x)$:
Answer: $MP(x) = -8x + 48$

b. Calculate and interpret the meaning of $MP(5)$.

Substitute $x = 5$ into the marginal profit function:

$$MP(5) = -8(5) + 48$$

$$MP(5) = -40 + 48 = 8$$

Interpretation: At a production level of 5 units per month, the profit is increasing at a rate of **$8 per unit**. This represents the approximate additional profit gained by producing and selling the **6th unit**.

c. How many units should the company produce to maximize the profit?

Step 1: Find $P'(x)$
(From part a):

$$P'(x) = -8x + 48$$

Step 2: Solve $P'(x) = 0$
Set the marginal profit to zero to find the value of x:

$$-8x + 48 = 0$$

$$8x = 48$$

$x = 6$

Step 3: Verify that $P''(x) < 0$
Find the second derivative to confirm that $x = 6$ results in a maximum:

$$P''(x) = -8$$

Since $-8 < 0$, the profit function results in a **maximum** at this point.

Answer: To maximize profit, the company should produce **6 units** each month.